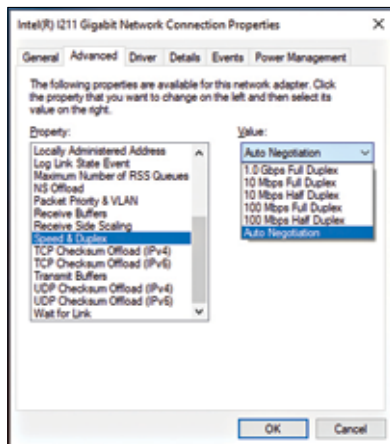


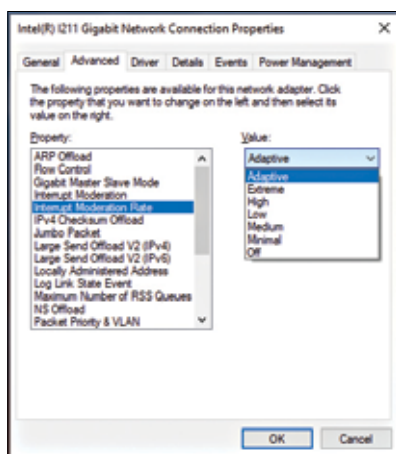
the dropdown list. Otherwise, the maximum speed visible would be 100 Mbps. Besides the speed, there options: Full

packets. Setting the rate at high or low has its own advantages and disadvantages. At a higher rate, the

CPU utilisation drops to give you better machine performance at the cost of less responsive packet handling. At a lower rate, the CPU utilisation increases, affecting machine performance but the system is now more responsive to packet handling. Depending on your network adapter, you might additionally get the option to change the interrupt moderation rate instead of only enabling or disabling it. Directly disabling interrupt moderation will give you the lowest possible latency. Of course, this will be bundled with a lower machine performance. For all the gamers out there, if you're playing a multiplayer game and happen



Pick the highest speed and favourable duplex



Lower is better

duplex (two-way simultaneous communication) or half-duplex (one-way communication). There's also the third option called auto-negotiation which gives the adapter the liberty to choose the highest speed and this will usually be set as default. We would recommend you to set the highest possible speed in the full duplex mode.

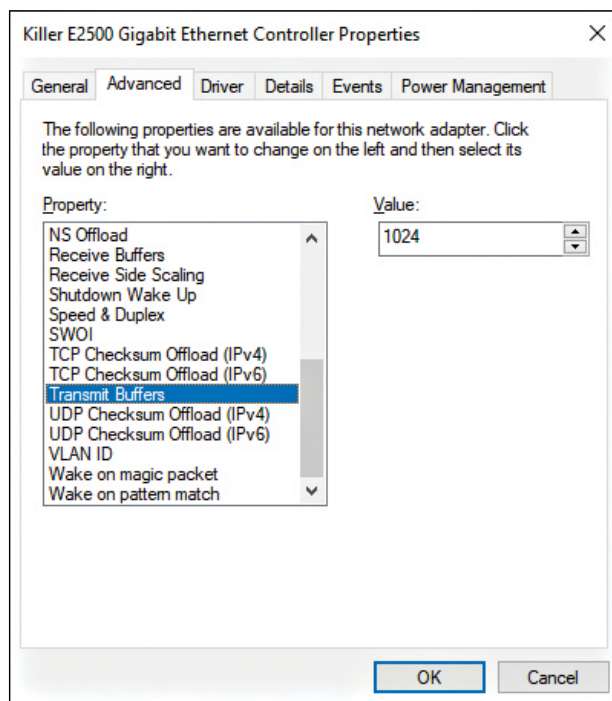
DISABLE INTERRUPT MODERATION

The interrupt moderation rate will determine how often the adapter interrupts the system to manage all the incoming and outgoing

to have a powerful CPU, you should definitely disable interrupt moderation.

ASSIGN HIGHER RECEIVE/TRANSMIT BUFFERS

When you're sending and receiving packets, there's a buffer of the system memory that is used by the adapter. Higher the buffer size, higher the performance. The only problem is that if you increase the buffer size, the memory footprint increases. Depending on your system, the default will be set automatically to the optimal size which is usually 256. Look for 'Receive Buffer' and 'Transmit



Transmit Buffer size should be twice that of Receive Buffer

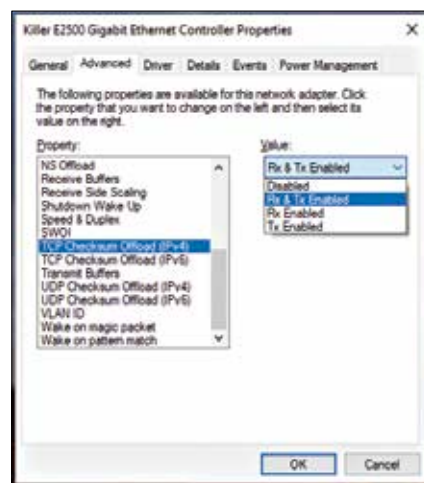
Buffer' in the list. The recommended size is either 512 or maybe even 1024 - you can go up to 2048 if you have enough RAM to support it.

Ensure that the transmit buffer is double the receive buffer. Just remember, that going below 256 will lead to dropped packets. Make sure you don't enter a huge buffer size, and keep testing until you achieve the best size based on your system memory.

DISABLE ALL TCP OFFLOADING SETTINGS

TCP offloading needs to be supported by the adapter in order for it to work. It will simply offload all the processing over to the network adapter instead of the CPU. This demands a fast and efficient network adapter, otherwise, it won't work the

way it is intended to. There have been times when the implementation of TCP offloading techniques have been buggy, so you need to ensure



Offload all processing to the network adapter

you have a good quality adapter. Disabling all TCP offloading settings will improve the throughput and free up the processor for other tasks. Disable 'Large Send Offload (IPv4)', 'Large Send Offload V2 (IPv4)', 'Large Send Offload V2 (IPv6)' and all the checksum offloads including IPv4/TCP/UDP. 